

Project info

1. Project title: Implementing and Evaluating Monte Carlo Tree Search for Connect Four
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Goals

3. Original Goals: Earlier ideas involved exploring larger-scale game environments and deep reinforcement learning approaches.
4. Current Goals: The goals remain largely consistent with the proposal's outline, but the focus has been narrowed to Connect Four. *Primary*: Implement a Monte Carlo Tree Search (MCTS) reinforcement learning framework for Connect Four and evaluate its performance under different computation constraints (rollout length and simulation count). *Secondary/Backup*: Compare MCTS performance with simpler reinforcement algorithms like temporal difference learning. *Stretch*: Compare with alternative algorithms such as double q network.
5. Reason for Change: Initial project scoping led to the narrowing of the project from larger-scale environments to Connect Four. This was done to balance project complexity with the available timeline, allowing more time for implementing, evaluating, and analyzing MCTS behavior.

Current status

Current work has focused primarily on literature review and project planning. Relevant literature reviewed includes papers on temporal-difference learning, MCTS, AlphaGo Zero, and Alpha Zero, with an emphasis on how MCTS with Reinforcement Learning (RL) can be applied to improve performance in turn-based games. The project is currently proceeding well with the narrowed scope, which has streamlined initial architectural decisions.

Preliminary planning is complete for two critical components. First, the environment will be integrated using Gymnasium-compatible interfaces. Second, evaluation metrics have been planned based on key performance indicators: win rate, simulation count, and computational cost. The immediate focus is now shifting to implementing a working Connect Four environment and establishing a baseline MCTS framework before moving on to comparing alternative algorithms and performance optimizations. No major approaches have been tried and failed; the narrowing of the scope from larger-scale games was a successful preemptive measure to manage the project timeline.

GitHub Link: https://github.com/falconeaj1/ME_569_RL

Timeline:

Week 1: Implement game and state representation. Integrate game with gymnasium and design baseline agents (random strategy). Review and prototype MCTS algorithms for smaller games, like tic-tac-toe, for debugging and validation.

Week 2: Implement core MCTS algorithm: selection, expansion, rollout/simulation, and backpropagation. Implement Upper Confidence Trees (UCT) selection rule. Develop an evaluation framework for automated self play and agent comparison.

Week 3: Run experiments varying rollout count, rollout policy, exploration constant. Compare MCTS against baseline agents. Potentially compare against other algorithms like temporal difference or heuristic guided rollouts.

Week 4: Perform final experiments and metric collections. Generate plots and analysis and create a final report. Update and clean code to make data processing pipeline clear.